Azure AI

Azure AI Vision

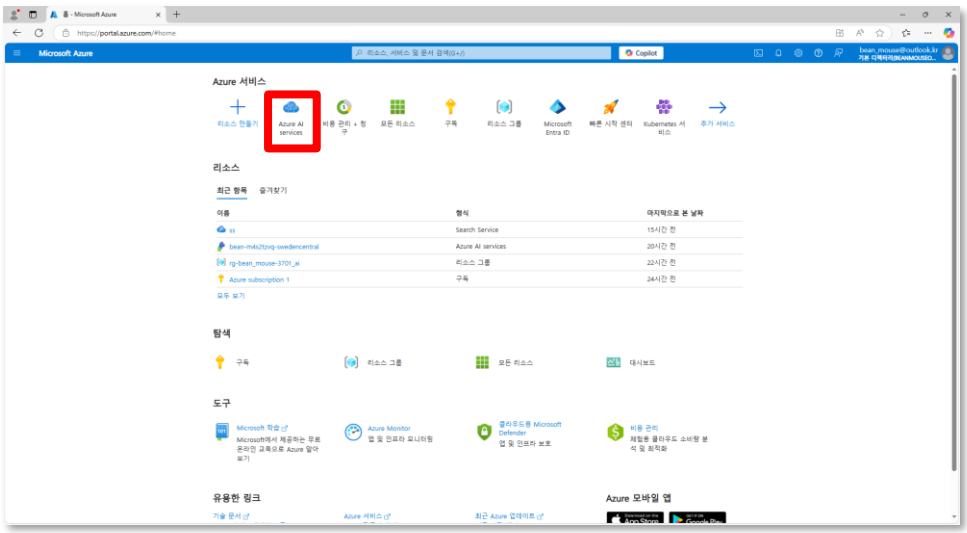
권기웅

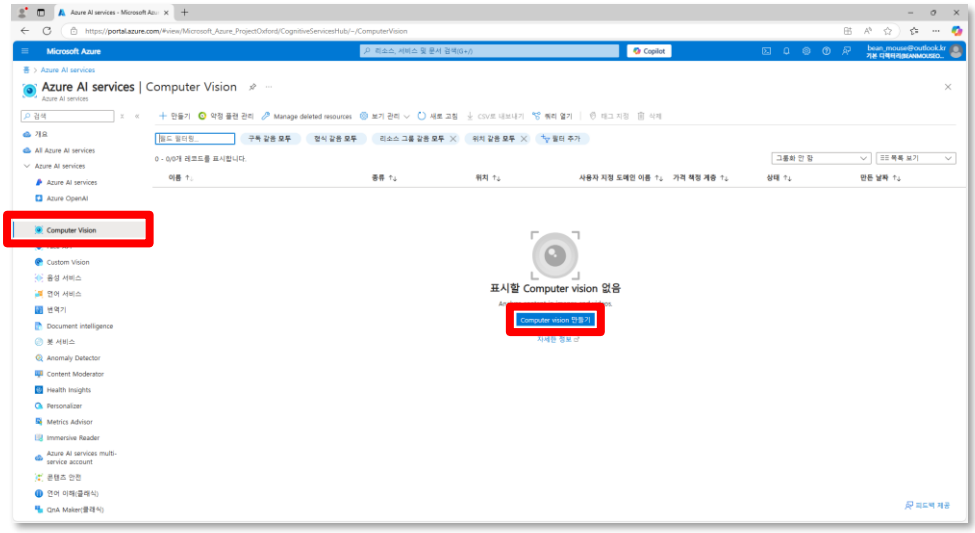
1. 환경설정

1.1 Azure AI Vision 리소스

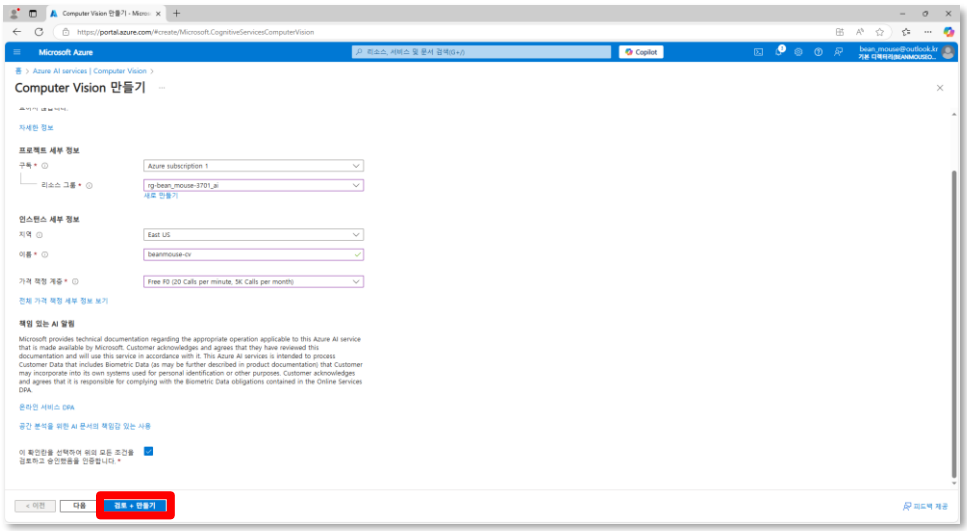
1.1.1 리소스 만들고

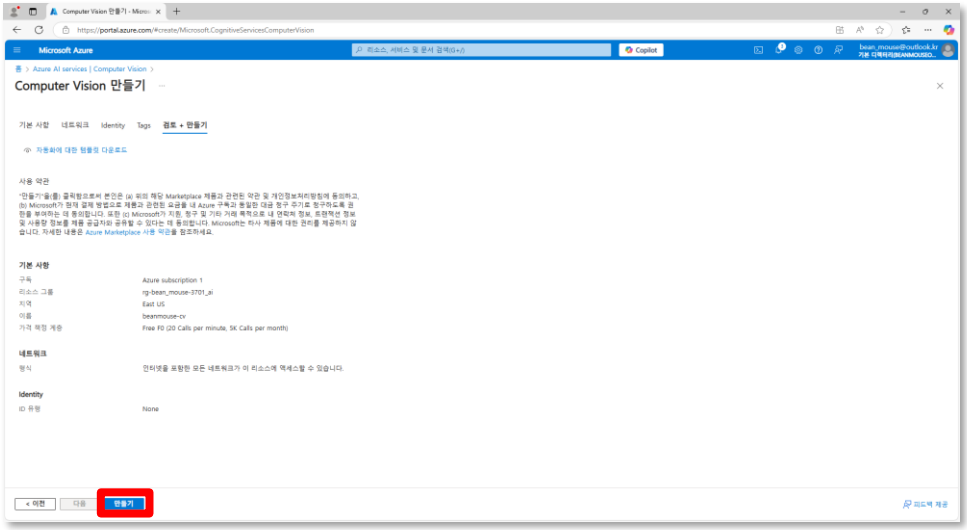
https://portal.azure.com 에서



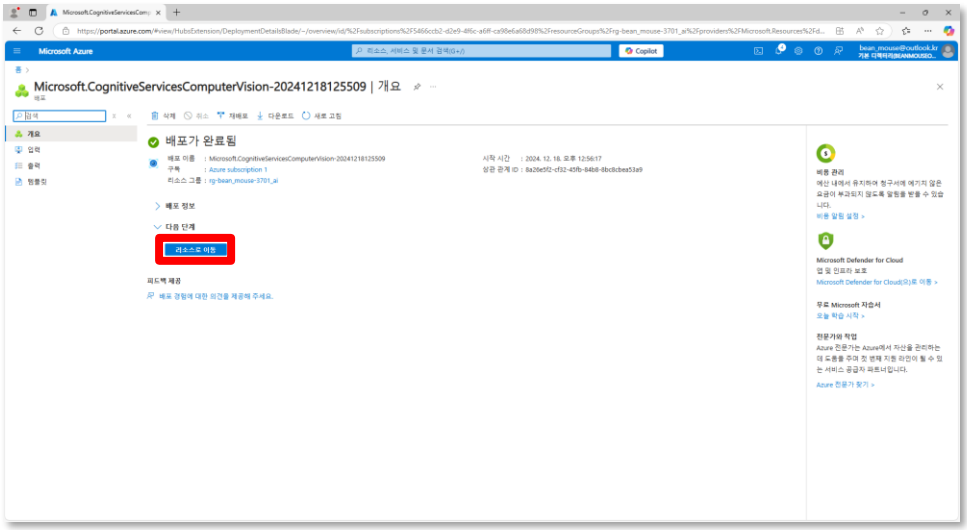


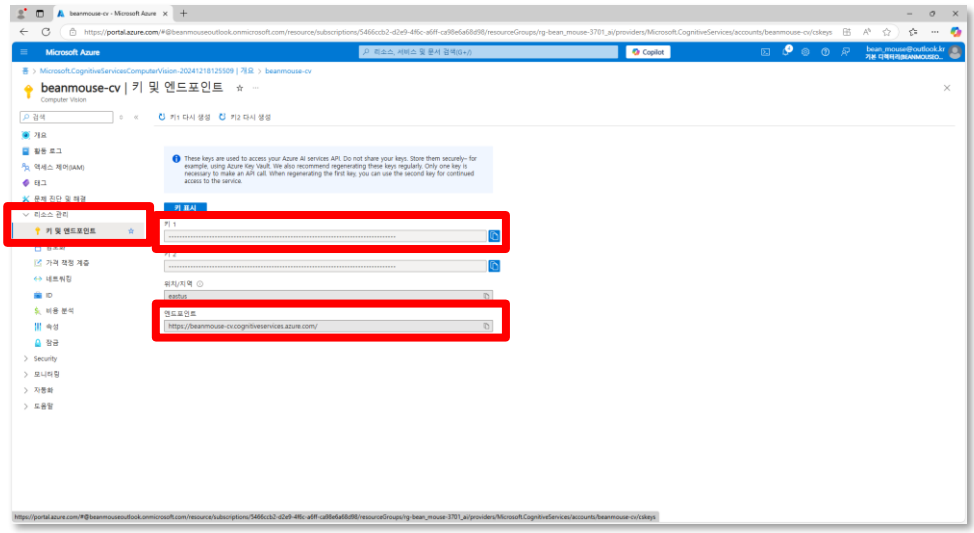
East US 2 안됨


Computer Vision 만들기
Azure subscription 1
rg-bean_mouse-3701_ai
East US
beanmouse-cv
Free F0 (20 Calls per minute, 5K Calls per month)
검토 + 만들기


Computer Vision 만들기
기본 사항
네트워크
Identity
Tags
검토 + 만들기
Azure subscription 1
rg-bean_mouse-3701_ai
East US
beanmouse-cv
Free F0 (20 Calls per minute, 5K Calls per month)
None
만들기

1.2 키, 엔드포인트 챙기기





2. OCR

2.1 Python 라이브러리 설치

2.1.1 Azure AI Vision 라이브러리

```
pip install --upgrade azure-cognitiveservices-vision-computervision
```

```
명령 프롬프트
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\soldesk>pip install --upgrade azure-cognitiveservices-vision-computervision
Collecting azure-cognitiveservices-vision-computervision
  Downloading azure_cognitiveservices_vision_computervision-0.9.1-py2.py3-none-any.whl (
36 kB)
Collecting msrest>=0.6.21
  Downloading msrest-0.7.1-py3-none-any.whl (85 kB)
     ---------------------------------------- 85.4/85.4 KB ? eta 0:00:00
Collecting azure-common~=1.1
```

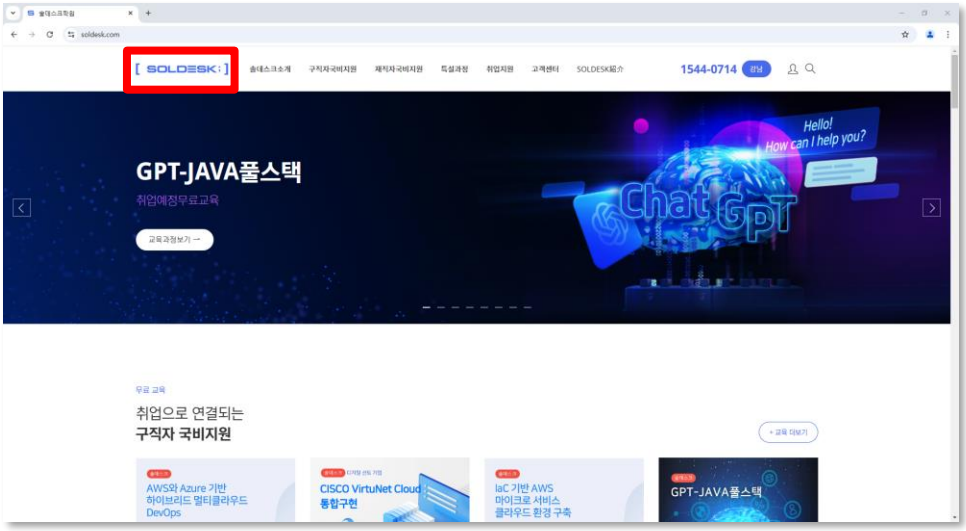
2.1.2 기타 라이브러리

```
pip install pillow
```

```
명령 프롬프트
C:\Users\soldesk>pip install pillow
Collecting pillow
  Using cached pillow-11.0.0-cp39-cp39-win_amd64.whl (2.6 MB)
Installing collected packages: pillow
Successfully installed pillow-11.0.0
WARNING: You are using pip version 22.0.4; however, version 24.3.1 is available.
You should consider upgrading via the 'C:\Users\soldesk\AppData\Local\Programs\Python\Py
thon39\python.exe -m pip install --upgrade pip' command.

C:\Users\soldesk>_
```

2.2 웹 이미지

2.2.1 웹 이미지 구해서



2.2.2 Python 프로그램

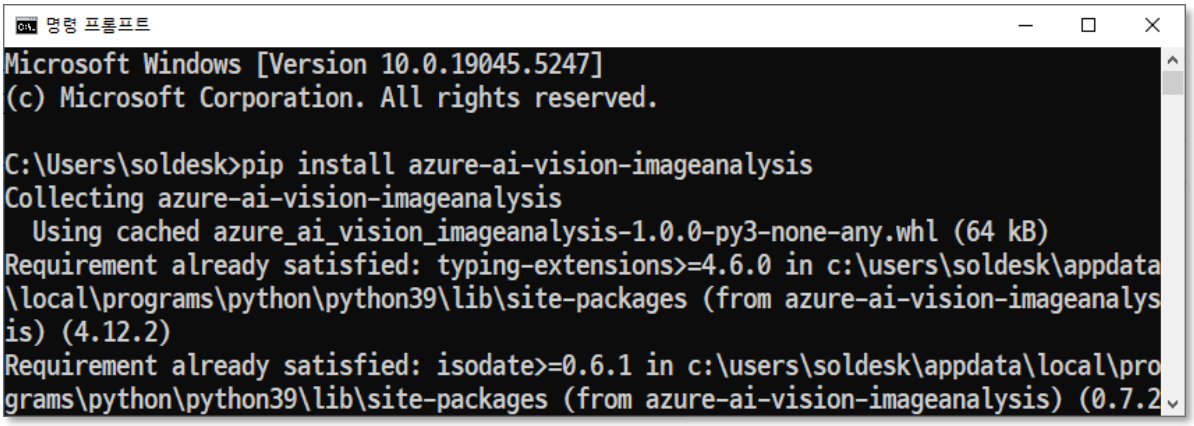
```python
# -*- coding:utf-8 -*-
from azure.cognitiveservices.vision.computervision import ComputerVisionClient
from azure.cognitiveservices.vision.computervision.models import OperationStatusCodes
from msrest.authentication import CognitiveServicesCredentials

import time

subscription_key = "8wOwb893nrWIfM2W2ltXyFQ3gc4DpDvSuEmWZIYop92oSUck48BnJQQJ99ALACYeBjFXJ3w3AAAFACOGdPWP"  # API키
endpoint = "https://beanmouse-cv.cognitiveservices.azure.com/"  # 엔드포인트

computervision_client = ComputerVisionClient(
    endpoint, CognitiveServicesCredentials(subscription_key)
)

read_image_url = "https://cdn.imweb.me/thumbnail/20230126/bfe794ac46524.png"

with open("D:/Kwon/자원/image/soldesk.jpg", "rb") as f:
    image_data = f.read()

read_response = computervision_client.read(read_image_url, raw=True)

read_operation_location = read_response.headers["Operation-Location"]
operation_id = read_operation_location.split("/")[-1]

while True:
    read_result = computervision_client.get_read_result(operation_id)
    if read_result.status not in ["notStarted", "running"]:
        break
    time.sleep(1)

if read_result.status == OperationStatusCodes.succeeded:
    for text_result in read_result.analyze_result.read_results:
        for line in text_result.lines:
            print(line.text)
            print(line.bounding_box)
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS
scode\extensions\ms-python.debugpy-2024.14.0-win32-x
py'
SOLDESK;
[64.0, 11.0, 508.0, 11.0, 508.0, 58.0, 64.0, 57.0]
```

3. 이미지 분석

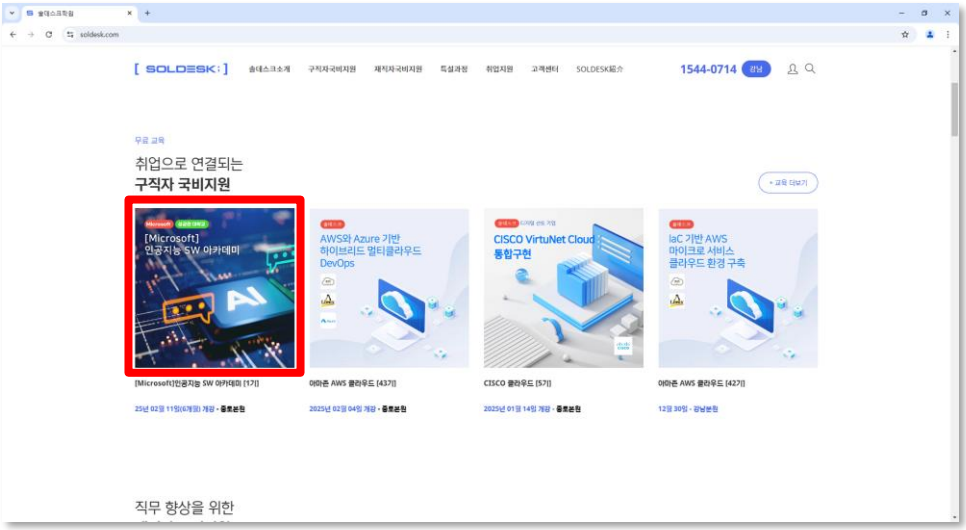
3.1 Python 라이브러리 설치

```
pip install azure-ai-vision-imageanalysis
```



3.2 웹 이미지

3.2.1 웹 이미지 구해서



3.2.2 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.vision.imageanalysis import ImageAnalysisClient
from azure.ai.vision.imageanalysis.models import VisualFeatures
from azure.core.credentials import AzureKeyCredential

endpoint = "https://beanmouse-cv.cognitiveservices.azure.com/"  # 엔드포인트
key = "8wOwb893nrWIfM2W2ltXyFQ3gc4DpDvSuEmWZIYop92oSUck48BnJQQJ99ALACYeBjFXJ3w3AAAFACOGdPWP"  # API키

client = ImageAnalysisClient(endpoint=endpoint, credential=AzureKeyCredential(key))

result = client.analyze_from_url(
    image_url="https://cdn-optimized.imweb.me/upload/S202205103bbf37d4d8229/957d4b0b552bc.png",
    visual_features=[VisualFeatures.CAPTION, VisualFeatures.READ],
)

print("Caption:")
if result.caption is not None:
    print("\t%s : %f" % (result.caption.text, result.caption.confidence))

print("Read:")
if result.read is not None:
    for line in result.read.blocks[0].lines:
        print("\t%s : " % line.text, line.bounding_polygon)
        for word in line.words:
            print("\t\t%s : %f" % (word.text, word.confidence), word.bounding_polygon)
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local\Programs\Python\Python39\python.exe'
scode\extensions\ms-python.debugpy-2024.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\launcher' '57526' '--' 'd:\K
on.py'
Caption:
        a computer chip with text overlay : 0.708664
Read:
        Microsoft :  [{'x': 68, 'y': 85}, {'x': 225, 'y': 85}, {'x': 225, 'y': 114}, {'x': 68, 'y': 114}]
                Microsoft : 0.994000 [{'x': 69, 'y': 86}, {'x': 224, 'y': 86}, {'x': 223, 'y': 115}, {'x': 68, 'y': 115}]
        성균관 대학교 :  [{'x': 265, 'y': 76}, {'x': 439, 'y': 77}, {'x': 439, 'y': 121}, {'x': 265, 'y': 120}]
                성 : 0.996000 [{'x': 272, 'y': 77}, {'x': 294, 'y': 77}, {'x': 294, 'y': 121}, {'x': 272, 'y': 121}]
                균 : 0.995000 [{'x': 295, 'y': 77}, {'x': 323, 'y': 77}, {'x': 323, 'y': 121}, {'x': 295, 'y': 121}]
                관 : 0.996000 [{'x': 324, 'y': 77}, {'x': 351, 'y': 77}, {'x': 351, 'y': 121}, {'x': 324, 'y': 121}]
                대 : 0.995000 [{'x': 352, 'y': 77}, {'x': 385, 'y': 78}, {'x': 385, 'y': 121}, {'x': 352, 'y': 121}]
                학 : 0.996000 [{'x': 386, 'y': 78}, {'x': 414, 'y': 78}, {'x': 414, 'y': 121}, {'x': 386, 'y': 121}]
                교 : 0.995000 [{'x': 414, 'y': 79}, {'x': 436, 'y': 79}, {'x': 436, 'y': 121}, {'x': 414, 'y': 121}]
        [Microsoft] :  [{'x': 61, 'y': 157}, {'x': 401, 'y': 157}, {'x': 401, 'y': 223}, {'x': 61, 'y': 223}]
                [Microsoft] : 0.989000 [{'x': 62, 'y': 159}, {'x': 399, 'y': 158}, {'x': 398, 'y': 224}, {'x': 62, 'y': 223}]
        인공지능 SW 아카데미 :  [{'x': 56, 'y': 230}, {'x': 654, 'y': 229}, {'x': 655, 'y': 296}, {'x': 56, 'y': 298}]
```

3.3 로컬 이미지

3.3.1 이미지 구해서

3.3.2 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.vision.imageanalysis import ImageAnalysisClient
from azure.ai.vision.imageanalysis.models import VisualFeatures
from azure.core.credentials import AzureKeyCredential

endpoint = "https://beanmouse-cv.cognitiveservices.azure.com/"
key = "8wOwb893nrWIfM2W2ltXyFQ3gc4DpDvSuEmWZIYop92oSUck48BnJQQJ99ALACYeBjFXJ3w3AAAFACOGdPWP"

client = ImageAnalysisClient(endpoint=endpoint, credential=AzureKeyCredential(key))

f = open("D:/Kwon/자원/image/20241218_131528.jpg", "rb")
image_data = f.read()

result = client.analyze(
    image_data=image_data,
    visual_features=[VisualFeatures.CAPTION, VisualFeatures.READ],
)

print("Caption:")
if result.caption is not None:
    print("\t%s : %f" % (result.caption.text, result.caption.confidence))

print("Read:")
if result.read is not None:
    for line in result.read.blocks[0].lines:
        print("\t%s : " % line.text, line.bounding_polygon)
        for word in line.words:
            print("\t\t%s : %f" % (word.text, word.confidence), word.bounding_polygon)
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local\Programs\Python\Python39\python.exe' 'c
scode\extensions\ms-python.debugpy-2024.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\launcher' '57675' '--' 'd:\Kwo
on.py'
Caption:
        a chair next to a fire extinguisher : 0.970137
Read:
        소화기 :  [{'x': 496, 'y': 91}, {'x': 554, 'y': 74}, {'x': 559, 'y': 89}, {'x': 500, 'y': 106}]
                소 : 0.995000 [{'x': 497, 'y': 91}, {'x': 508, 'y': 87}, {'x': 512, 'y': 103}, {'x': 501, 'y': 106}]
                화 : 0.991000 [{'x': 517, 'y': 85}, {'x': 526, 'y': 82}, {'x': 530, 'y': 98}, {'x': 521, 'y': 100}]
                기 : 0.994000 [{'x': 537, 'y': 79}, {'x': 546, 'y': 77}, {'x': 551, 'y': 92}, {'x': 541, 'y': 95}]
        FAT CANNINGITISACR :  [{'x': 501, 'y': 107}, {'x': 553, 'y': 91}, {'x': 555, 'y': 97}, {'x': 502, 'y': 113}]
                FAT : 0.113000 [{'x': 501, 'y': 108}, {'x': 513, 'y': 105}, {'x': 515, 'y': 110}, {'x': 504, 'y': 113}]
                CANNINGITISACR : 0.009000 [{'x': 513, 'y': 105}, {'x': 551, 'y': 92}, {'x': 554, 'y': 98}, {'x': 516, 'y': 110}]
        권기응 :  [{'x': 386, 'y': 377}, {'x': 417, 'y': 365}, {'x': 422, 'y': 377}, {'x': 391, 'y': 391}]
                권 : 0.994000 [{'x': 387, 'y': 376}, {'x': 396, 'y': 372}, {'x': 402, 'y': 386}, {'x': 393, 'y': 390}]
                기 : 0.996000 [{'x': 397, 'y': 372}, {'x': 406, 'y': 369}, {'x': 411, 'y': 382}, {'x': 402, 'y': 386}]
                응 : 0.763000 [{'x': 408, 'y': 368}, {'x': 416, 'y': 365}, {'x': 422, 'y': 378}, {'x': 413, 'y': 382}]
        더소화기 :  [{'x': 545, 'y': 374}, {'x': 610, 'y': 365}, {'x': 613, 'y': 381}, {'x': 547, 'y': 392}]
                더 : 0.650000 [{'x': 545, 'y': 375}, {'x': 554, 'y': 374}, {'x': 557, 'y': 391}, {'x': 547, 'y': 393}]
```

3.3.3 객체, 사람, 태그

```python
# -*- coding:utf-8 -*-
from azure.ai.vision.imageanalysis import ImageAnalysisClient
from azure.ai.vision.imageanalysis.models import VisualFeatures
from azure.core.credentials import AzureKeyCredential

endpoint = "https://beanmouse-cv.cognitiveservices.azure.com/"
key = "8wOwb893nrWIfM2W2ltXyFQ3gc4DpDvSuEmWZIYop92oSUck48BnJQQJ99ALACYeBjFXJ3w3AAAFACOGdPWP"

client = ImageAnalysisClient(endpoint=endpoint, credential=AzureKeyCredential(key))

f = open("D:/Kwon/자원/image/20241218_131528.jpg", "rb")
image_data = f.read()

result = client.analyze(
    image_data=image_data,
    visual_features=[VisualFeatures.CAPTION, VisualFeatures.READ, VisualFeatures.OBJECTS, VisualFeatures.PEOPLE, VisualFeatures.TAGS],
)

print("Caption:")
if result.caption is not None:
    print("\t%s : %f" % (result.caption.text, result.caption.confidence))

print("Read:")
if result.read is not None:
    for line in result.read.blocks[0].lines:
        print("\t%s : " % line.text, line.bounding_polygon)
        for word in line.words:
            print("\t\t%s : %f" % (word.text, word.confidence), word.bounding_polygon)

print("Objects:")
if result.objects is not None:
    print(result.objects)

print("People:")
if result.people is not None:
    print(result.people)

print("Tags:")
if result.tags is not None:
    print(result.tags)
```

```
Objects:
{'values': [{'boundingBox': {'x': 188, 'y': 50, 'w': 335, 'h': 486}, 'tags': [{'name': 'chair', 'confidence': 0.797}]}]}
People:
{'values': []}
Tags:
{'values': [{'name': 'floor', 'confidence': 0.9678898453712463}, {'name': 'furniture', 'confidence': 0.9214739799499512},
ence': 0.8952429294586182}, {'name': 'chair', 'confidence': 0.8893395662307739}, {'name': 'indoor', 'confidence': 0.82078
round', 'confidence': 0.7873995304107666}]}
```